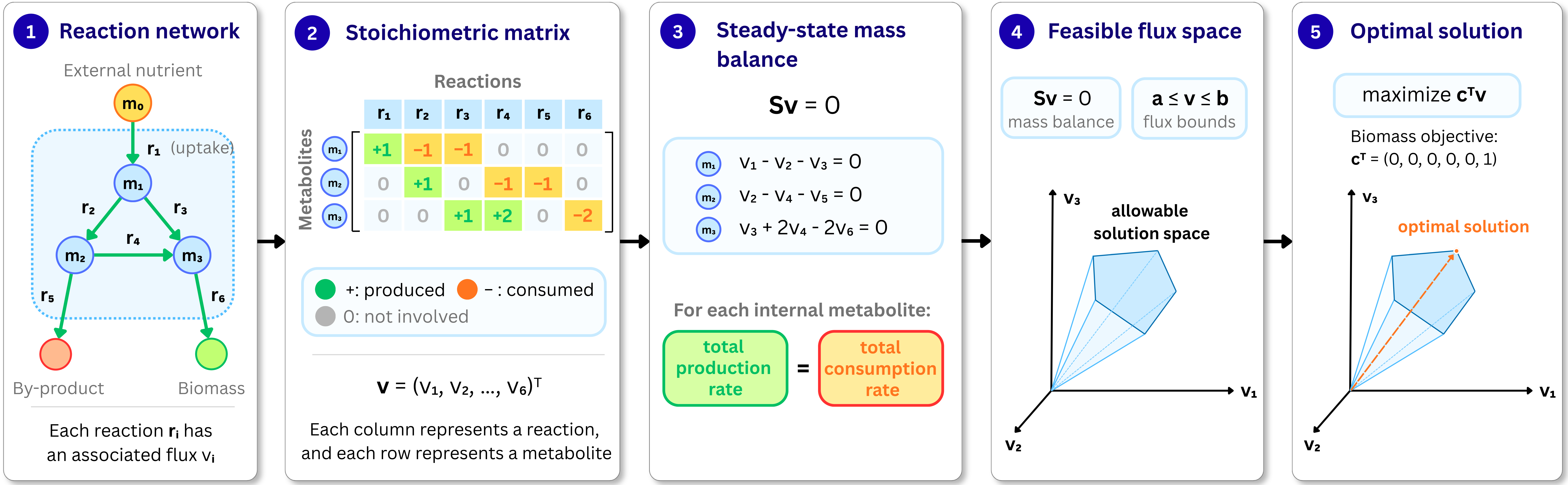
1 Reaction network
External nutrient
m_0
r_1 (uptake)
m_1
r_2
r_3
r_4
m_2
m_3
r_5
r_6
By-product
Biomass
Each reaction r_i has an associated flux v_i
2 Stoichiometric matrix
Reactions
Metabolites
r_1 r_2 r_3 r_4 r_5 r_6
m_1 +1 −1 −1 0 0 0
m_2 0 +1 0 −1 −1 0
m_3 0 0 +1 +2 0 −2
+: produced
− : consumed
0: not involved
$\mathbf{v} = (v_1, v_2, ..., v_6)^T$
Each column represents a reaction, and each row represents a metabolite
3 Steady-state mass balance
$\mathbf{Sv} = 0$
m_1 $v_1 - v_2 - v_3 = 0$
m_2 $v_2 - v_4 - v_5 = 0$
m_3 $v_3 + 2v_4 - 2v_6 = 0$
For each internal metabolite:
total production rate = total consumption rate
4 Feasible flux space
$\mathbf{Sv} = 0$
mass balance
$\mathbf{a} \leq \mathbf{v} \leq \mathbf{b}$
flux bounds
v_3
allowable solution space
v_1
v_2
5 Optimal solution
maximize $\mathbf{c}^T\mathbf{v}$
Biomass objective:
$\mathbf{c}^T = (0, 0, 0, 0, 0, 1)$
v_3
optimal solution
v_1
v_2